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Modeling downdraft biomass gasification process by restricting chemical reaction equilibrium with Aspen Plus

Jun Han[a], Yan Liang[a], Jin Hu[b], Linbo Qin[a], Jason Street[b], Yongwu Lu[b], Fei Yu[b,*]

[a] Hubei Key Laboratory for Efficient Utilization and Agglomeration of Metallurgic Mineral Resources, Wuhan University of Science and Technology, Wuhan 430081, PR China

[b] Department of Agricultural and Biological Engineering, Mississippi State University, MS 39762, USA

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ABSTRACT

Thermo-chemical conversion of biomass has been regarded as one of attractive routes of producing the clean and environmental friendly bio-fuels. Downdraft biomass gasification process has been served as a key technology to provide the bio-syngas with high quality, which can be used to produce the renewable liquid transportation fuels through Fischer Tropsch Synthesis in Mississippi State University. In order to provide the performance data of the integrated Biomass to Liquid system and future process optimization, a comprehensive model of the downdraft biomass gasification process based on Aspen Plus by minimizing Gibbs free energy with restricting chemical reaction equilibrium in the gasification reduction zone has been developed. The model was successfully validated with the experimental data from the hardwood chips gasification. Sensitivity analysis was also performed to investigate the effects of gasification temperature, equivalence ratio, and biomass moisture content on the quality of bio-syngas. All the investigated factors have been found with a significant effect.

1. Introduction

Biofuel production from biomass thermo-chemical conversion has been regarded as one of attractive routes to provide alternative and clean fuels [1,2]. An integrated continuous process of Biomass to Liquid via Fischer Tropsch, involving biomass gasification, bio-syngas cleaning, and Fischer-Tropsch Synthesis, has been established in Mississippi State University to produce gasoline [3–6], Kerosene [7] and also mixed alcohols [8] from biomass. The downdraft biomass gasification in the integrated process serves as a key technology to provide the high quality bio-syngas for the downstream catalytic conversion process [4].

Biomass gasification is a thermo-chemical process of converting carbonaceous biomass into gas mixture, majorly hydrogen, carbon monoxide, carbon dioxide and methane, with some tar, char, ash, and also trace amount of oxygen, sulfide and nitrogen containing compounds [9]. Gasification process can be influenced by gasifier configuration and the feedstock used in the reaction. However, the major process and chemical reactions in downdraft biomass gasification are largely similar and can be divided into four physico-chemical processes, including drying, pyrolysis, oxidation and reduction [10]. The chemical reactions occurred in the biomass gasification have been summarized and presented in Table 1.

Biomass is usually pre-treated (or pre-dried) to reduce the moisture content to the desired degree before it is feed to the gasification reactor. Within downdraft gasification reactor, biomass is firstly dried at around 100 °C to vaporize the remaining water. The steam from the drying process may be involved into the subsequent chemical reactions or be mixed with the syngas as moisture. The dried biomass will then undergo the pyrolysis or devolatilization process. In this stage, biomass will be broken down into medium molecules and solid carbon (char) from the large molecules (majorly cellulose, hemi-cellulose, and lignin). Depending on the residence time during the reaction, the medium molecules will continue decomposing into the small molecules (R1), such as hydrogen, carbon monoxide, carbon dioxide, methane, etc. if residence time is long enough. If not, the medium molecules will form condensates (such as tars and oils) and enter into the oxidation zone. Gasification agent, usually ambient air, oxygen, and steam, is introduced into the oxidation zone. Products from the pyrolysis zone will react with the oxidation agent to produce smaller molecules (R2 to R6). In the reduction zone, water gas shift reaction (R7), Boudouard reaction (R8), water gas reaction (R9), methanation reaction (R10) and steam methane reforming (R11) will occur due to insufficient oxygen in the high temperature zone. The hot gas, char, tar and ash are then conducted out of the gasification reactor for the post-gasification treatment such as separation, cooling and gas drying [10–13].

* Corresponding author.
E-mail address: fyu@abe.msstate.edu (F. Yu).

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Table 1
Chemical reactions in the downdraft biomass gasification process.

ID	Chemical Reaction	Reaction Heat	Reaction Name	Zone[a]
R1	Dried Biomass → Volatiles (H_2, CO, CO_2, H_2O, CH_4 Equivalent and Tar) + Char	NA[b]	Pyrolysis	Pyrolysis
R2	$H_2 + 0.5O_2 \rightarrow H_2O$	(−242 MJ/kmol)	H_2 Oxidation	Oxidation
R3	$CO + 0.5O_2 \rightarrow CO_2$	(−283 MJ/kmol)	CO Oxidation	Oxidation
R4	$C_{1.16}H_4$[c] $+ 1.58O_2 \rightarrow 1.16CO + 2H_2O$	NA	Light Hydrocarbon Oxidation	Oxidation
R5	$C_6H_{6.2}O_{1.2}$[d] $+ 4.45O_2 \rightarrow 6CO + 3.1H_2O$	NA	Heavy Hydrocarbon Oxidation	Oxidation
R6	$C + 0.5O_2 \rightarrow CO$	(−111 MJ/kmol)	Char Partial Oxidation	Oxidation
R7	$CO + H_2O \leftrightarrow CO_2 + H_2$	(−41 MJ/kmol)	Water Gas Shift	Reduction
R8	$C + CO_2 \leftrightarrow 2CO$	(+172 MJ/kmol)	Boudouard	Reduction
R9	$C + H_2O \leftrightarrow CO + H_2$	(+131 MJ/kmol)	Water Gas	Reduction
R10	$C + 2H_2 \leftrightarrow CH_4$	(−75 MJ/kmol)	Methanation	Reduction
R11	$CH_4 + H_2O \leftrightarrow CO + 3H_2$	(+206 MJ/kmol)	Steam Methane Reforming	Reduction
R12	$H_2 + S = H_2S$	NA	H_2S Formation	NA
R13	$0.5 N_2 + 1.5H_2 \leftrightarrow NH_3$	NA	NH_3 Formation	NA

[a] Indicates reaction zones in the gasifier.
[b] NA = Not Available.
[c] $C_{1.16}H_4$ indicates light hydrocarbon or methane equivalent.
[d] $C_6H_{6.2}O_{1.2}$ indicates heavy hydrocarbon, such as tar.

It was reported that the quality of bio-syngas, such as gas composition and hydrogen to carbon monoxide ratio, from the biomass gasification will vary with different gasification operation conditions and feedstock properties [14,15]. Meanwhile, the bio-syngas quality has an important effect on catalytic synthesis in the downstream process for liquid fuel production [16]. Therefore, it was necessary to study the influence of the operation parameters of the downdraft biomass gasifier on bio-syngas quality, which can provide the performance data for the integrated process operation and further process optimization. Aspen Plus has already been used in many research areas, such as waste incineration [17] and hydrogen production [18,19]. Frey et al. have successfully improved system integration for Integrated Gasification Combined Cycle (IGCC) systems by using Aspen Plus [20]. Process simulation modeling using Aspen Plus is seem to be a useful tool to evaluate and optimize technology options. However, few researchers reported the chemical reactions such as R4 and R5 in the simulation, which would generate CO due to the partial oxidation [13,21].

The primary objective of this research was to develop a comprehensive model for the downdraft biomass gasification process in Mississippi State University with the aid of commercial process simulation software Aspen Plus. The model was validated with the experimental data from the hardwood chips gasification. The sensitivity analysis was also performed to investigate the effect of gasification temperature, equivalence ratio and biomass moisture content on the quality of bio-syngas from downdraft biomass gasifier.

2. Methodology

2.1. Gasification system

The pilot-scale downdraft biomass gasification system was purchased from CPC (Community Power Corp., Littleton, Colo.) and installed in the Pace Seed Lab of Department of Agricultural and Biological Engineering, Mississippi State University. The system has successfully performed several running of biomass gasification. The gasifier system includes drying/storage bins with capacities of about 250 kg, a conveyor for distribution of the chips onto a vibratory size classifier, and a screw-auger for transport of the chips into the gasifier. The gasifier is cylindrical with a diameter of 35.6 cm and length 118.3 cm with five thermocouples and secondary air inlets to control the temperature of the reaction located at 20.2, 35.3, 46.8, 60.2 and 73.5 cm from the bottom. The unit operates continuously and is open to the atmosphere. The feedstock is initially ignited by a resistance heater and the downdraft is created by a roots blower. In this work, the flow rate was set at 65 m^3/h (where a m^3 is defined at 0 °C and 101.3 kPa). The producer gas exits the gasifier at a temperature of about 500 °C and undergoes secondary cooling to a temperature of about 110 °C with an air cooled, shell and tube heat exchanger. The cooled gas is filtered to remove particulates, and analyzed in real time for oxygen, carbon monoxide, carbon dioxide, hydrogen and methane using a Nova 7900P5 infrared gas analyzer.

2.2. Modeling and simulation software

In this study, Aspen Plus V7.3 was purchased from Aspen Technology (Burlington, Massachusetts) for modeling the downdraft biomass gasification process. Aspen Plus is a steady state chemical process simulator. It can be used in most computational platform nowadays. User can build the process flowsheet by using unit operation blocks in the software, and then specify component information, physical property method, and stream information for calculation. Aspen Plus has extensive built-in physical properties database that can be used in the simulation calculation, it can solve the process module by module sequentially, and calculate outlet stream information through inlet stream properties for each block. In order to simulate non-conventional components, such as biomass, coal, and municipal solid waste, external FORTRAN code can also be incorporated into the Aspen Plus model for the non-conventional process calculation (as supplemental materials).

2.3. Physical property method

The IDEAL property method (Ideal gas and Raoult's Law) has been used in this simulation, since the process involves the conventional components at low pressure. Biomass and ash were defined as the non-conventional components in this process, the enthalpy and density model used for both biomass and ash were HCOALGEN and DCOALIGT. The stream class had been defined as MCINCPSD, since there were mixtures of the conventional and non-conventional streams with solid particles in the process.

2.4. Model assumptions

The main assumptions in the downdraft biomass gasification are [22]

Steady state and isothermal

- Tar and other heavy hydrocarbons are not considered
- Char is composed of carbon and ash
- Sulphur reacts to form H_2S only
- NH_3 formed only, with no nitrogen oxides considered

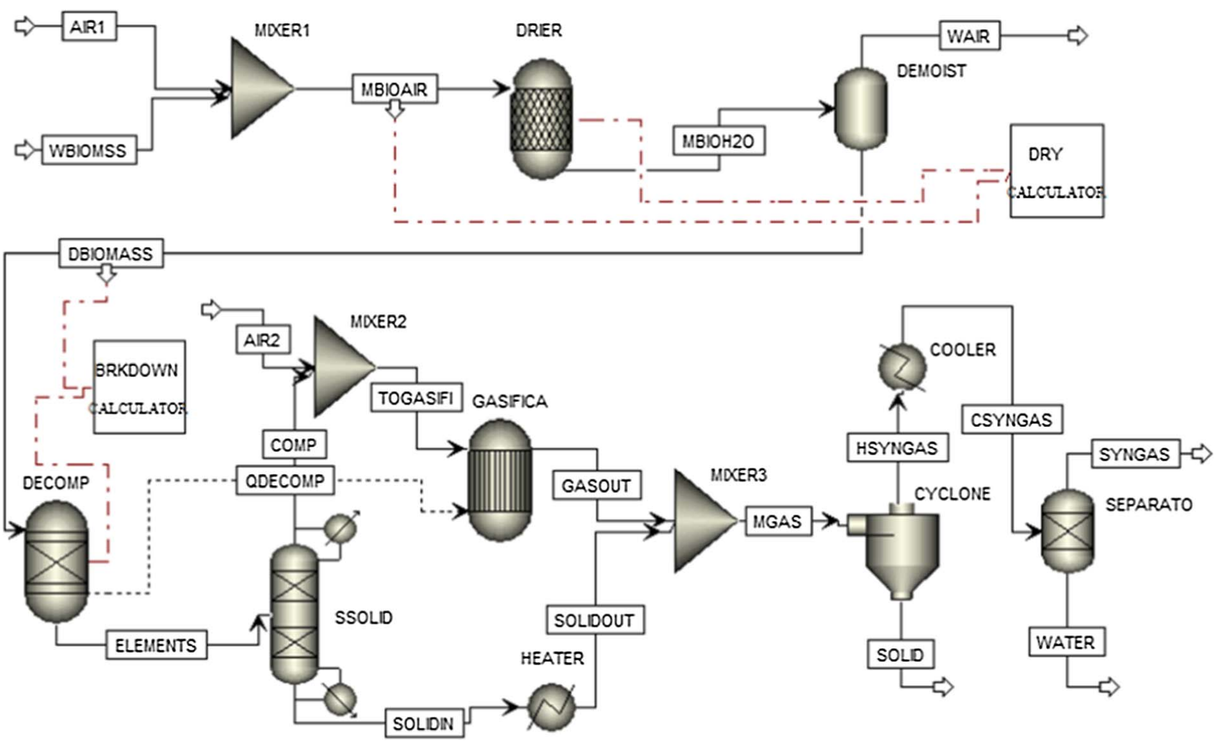

Fig. 1. Aspen Plus flowsheet for the comprehensive model of downdraft biomass gasification process.

3. Aspen Plus model development

The downdraft biomass gasification process has been modeled with three basic stages: biomass pre-drying, biomass gasification, and bio-syngas post-treatment. In the first stage, biomass moisture content will be reduced to the desired degree before loading into the biomass gasification reactor. In the second stage, the pre-dried biomass will be broken down into basic elements and then react with gasification agent to produce mixture of hot gases, solid particles, and moisture. In the third stage, the mixture from previous stage will be cooled, solids will be separated and moisture will be removed to produce bio-syngas.

3.1. Aspen Plus flowsheet

The downdraft biomass gasification flowsheet has been developed and been shown in Fig. 1. The solid lines indicate the stream directions. The black dash line indicates the energy flow. The red dash lines indicate that the block was controlled by user programed FORTRAN statement. The unit operation blocks used in the flowsheet have been summarized and presented in Table 2, with a brief description of default ID, which is unit operation block name defined by software developer, and assigned ID, which is created by user, and a short description for the function of each block.

3.2. Model description

The stream 'WBIOMASS' was defined as the original wet biomass before the pre-drying process, specified as the non-conventional component with the ultimate and proximate analysis input. It contains moisture content around 25% from the biomass provider without any treatment [23]. The wet biomass was generally pre-dried by forcing ambient air with fans before the gasification. In the model, 'WBIOMASS' was mixed with 'AIR1' and sent to 'DRIER' for pre-drying the wet biomass. The 'DRIER' was the RStoic block in Aspen Plus, it permitted the use of several reactions with the molar extent of conversion or fractional conversion of a component. The 'DRIER' was supplemented and controlled by external FORTRAN statement 'DRY' in the C-ALCULATOR block to reduce the moisture content to a predetermined point. Then the dried biomass and air with moisture were sent out through stream 'MBIOH2O' to 'DEMOSIT', then the 'DEMOIST' separated the dried biomass and wet air. 'WAIR' was released into the ambient environment and 'DBIOMASS' was sent to 'DECOMP' reactor.

Table 2
List of unit operation blocks utilized in Aspen Plus flowsheet.

Default ID	Assigned ID	Description
RStoic	DRIER	Reduce the moisture content of wet biomass by defining stoichiometric reaction
RYield	DECOMP	Decompose the non-conventional biomass into conventional components
RGibbs	GASIFICA	Simulate the reactions between conventional components through Gibbs free energy minimization and chemical equilibrium restriction
Flash2	DEMOIST	Separate the wet air from dried biomass
SSplit	CYCLONE	Separate solids from the hot gases
Mixer	MIXER1	Mix wet biomass feed with ambient air to reduce the moisture
	MIXER2	Mix elements from decomposition reactor with gasification agent ambient air
	MIXER3	Mix unreacted carbon with gas mixture from Gibbs reactor
Sep2	SSOLID	Separate a portion of carbon from the feed as unreacted carbon
	SEPARATO	Separate water from cool syngas to remove the moisture
Heater	HEATER	Heat up the unreacted carbon to Gibbs reactor temperature before mixing
	COOLER	Cool the hot syngas to ambient temperature to remove moisture

The 'DECOMP' was the RYield reactor block in Aspen Plus, it was used when the distribution of products was known while the reaction stoichiometry and kinetics were unknown. The 'DECOMP' reactor was used to simulate the decomposition of the dried biomass into its elemental components, including C, H_2, O_2, N_2, S, ASH, and moisture. This function was performed through user programed 'BRKDOWN' FORTRAN statement with the biomass ultimate analysis data. The enthalpy of the outlet stream 'ELEMENTS' was not equal to the enthalpy of the stream 'DBIOMASS' due to break up of chemical bonds, so the heat stream 'QDECOMP' was connect between 'DECOMP' and 'GASIFICA' to replenish the difference in enthalpy. 'SSOLID' block was used to separate a portion of carbon from the feed 'ELEMENTS' as the unreacted carbon. Carbon conversion rate for the downdraft biomass gasifier was reported from 96% to 99% [23]. The constituents of stream 'COMP' was then mixed with ambient air 'AIR2' and sent to 'GASIFICA' reactor. The 'GASIFICA' was RGibbs block in Aspen Plus, it can be used to establish the chemical equilibrium composition between reactants and products. It was useful when the temperature and pressure of the reactions were known while the reaction stoichiometry was unknown. The 'GASIFICA' reactor was used to simulate the oxidation and reduction zones of the downdraft gasifier by modeling chemical equilibrium through minimizing Gibbs free energy and restricting chemical reaction equilibrium in the gasification reduction zone (R7 to R11) to tune the product gas composition from the gasifier. The non-conventional component's Gibbs free energy could not be calculated by the simulator, however, it could be achieved by decomposing the biomass into its elements by 'DECOMP' reactor and then calculated in the 'GASIFICA' reactor. The unreacted carbon was then heated up to mix with the product stream 'GASOUT' from the 'GASIFICA' reactor to form 'MGAS' stream.

The block 'CYCLONE' was used to separate solids, such as char and ash from the gas mixture, and its efficiency of cyclone was 85%. The top outlet stream 'HSYNGAS' was composed of all the gases and moisture from the 'GASIFICA' reactor. It was then cooled down to ambient temperature by 'COOLER'. Moisture in 'HSYNGAS' was condensed and then separated by 'SEPARATO'. The top outlet stream 'SYNGAS' was the final gas from the biomass gasification.

3.3. Model validation

The model was validated with the experiments of Wei et al. [23], which were conducted on the CPC pilot-scale downdraft biomass gasifier using ambient air as gasification agent. The hardwood chips of red oak used in the experiments were purchased from the Weyerhauser Company (Columbus, Mississippi). The ultimate and proximate analyses of the hardwood chips can be found in Table 3.

In this study, Equivalence Ratio and H_2/CO Ratio were defined respectively as follows:

$$ER = \frac{weight\ oxygen(air)/weight\ dry\ biomass}{Stoichiometric\ oxygen(air)/biomass\ ratio} \tag{1}$$

$$\mathrm{H_2/CO}\ ratio = \frac{Volume\ percentage\ of\ hygrogen(dry\ basis)}{Volume\ percentage\ of\ carbon\ monoxide(dry\ basis)} \tag{2}$$

The lower heating value of the product gas depended on the volume percentage of CO, H_2 and CH_4 in the bio-syngas and could be calculated as follow:

$$LHV = X_{H_2}LHV_{H_2} + X_{CO}LHV_{CO} + X_{CH_4}LHV_{CH_4} \tag{3}$$

where X was the volume percentage of each gas component. The lower heating values of H_2, CO and CH_4 are 11.2 MJ/Nm^3, 13.1 MJ/Nm^3 and 37.1 MJ/Nm^3, respectively.

Simulation results were compared with all sets of experimental data. The sum squared deviation method was used to estimate the accuracy of simulation results [24].

Table 3
Ultimate and proximate analyses of feedstock (hardwood chips).[a]

Ultimate Analysis (dry basis)		
Carbon	wt.%	49.817
Hydrogen	wt.%	5.556
Oxygen	wt.%	43.425
Nitrogen	wt.%	0.078
Sulphur	wt.%	0.005
Ash	wt.%	1.119
TOTAL	wt.%	100
Proximate Analysis (dry basis)		
Volatile Matter	wt.%	79.850
Fixed Carbon	wt.%	19.031
Ash	wt.%	1.119
TOTAL	wt.%	100
Moisture (received basis)	wt.%	25
Moisture (After Pre-drying)	wt.%	8.910
HHV (dry basis)	MJ/kg	18.580
Bulk Density	kg/m^3	222.150

[a] Normalized from data reported by Wei et al. [8,16,17].

$$RSS = \sum_{i=1}^{N}\left(\frac{y_{ie}-y_{ip}}{y_{ie}}\right)^2 \tag{4}$$

$$MRSS = \frac{RSS}{N} \tag{5}$$

$$Mean\ \mathrm{error} = \sqrt{MRSS} \tag{6}$$

The gasification parameters and other experiment data had been entered into the model. Experimental run number 19 [23] was chosen for a detailed comparison and analysis, which was gasified at 775 °C. Table 4 compared the experimental run number 19 results with the model prediction results from this study. Moreover, the comparison of the predicted results by our model with the experimental results in other researchers were also conducted in Table 4.

The model predictions were in good agreement with the experimental data. For example, the syngas composition from the experiment was detected as 18.32% H_2, 20.93% CO, 12.87% CO_2, 44.79% N_2, 3.09% CH_4 and trace amount of NH_3, H_2S and O_2, the lower heating value of the bio-syngas from the experiment was 5.14 MJ/Nm^3. The model in this study predicted composition of 18.29% H_2, 21.31% CO and 11.36% CO_2 with very good accuracy. However, the concentration of CH_4 was underestimated, which was a quite common problem in the equilibrium modeling [25]. Null value for methane concentration was usually predicted in equilibrium modeling above 800 °C [26]. This underestimation can be explained by the difference exists between a real gasification system and an ideal reactor at chemical equilibrium [27]. The syngas from the gasifier generally contained char and tar, which was not considered in equilibrium models, and much more hydrocarbons (especially methane) than model predicted [22]. The predicted lower heating value was 4.86 MJ/Nm^3, which was lower than the experimental data because of the under-predicted CH_4 content. In order to solve the above problem, Gagliano et al. proposed a correlation between biomass moisture content and the equivalent ratio [28].

4. Sensitivity analysis and discussion

The biomass moisture content, the equivalent ratio (ER), and reaction temperature are the most important parameters that influence the chemical composition and LHV of the syngas [28]. Thus, a parametric study has been performed to predict the syngas composition, LHV varying the moisture content, ER and reaction temperature. The sum squared deviation method was used to estimate the accuracy of simulation results [24], the mean error of H_2, CO, and CO_2 were 0.045, 0.1472 and 0.1567.

Figs. 2–4 show the results of model sensitivity. The effect of

Table 4
Experimental data and model predictions.

	Experimental run #19	Our model	Nikoo [24] experiment	Nikoo model	Our model	Lopez [21] experiment	Lopez model	Our model
H_2	18.32	18.29	24	39	27	35–45	50	41
CO	20.93	21.31	45	46	50	22–25	17	27
CO_2	12.87	11.36	22	22	21	20–25	22	28
N_2	44.79	48.99						
CH_4	3.09	0.2	9	10	2	0–10	11	4
Mean error[a]		0.89		0.36	0.28		0.44	0.04
LHV (MJ/Nm^3)	5.14	4.86						

[a] without considering CH_4.

gasification temperature on the bio-syngas composition for hardwood chips is shown in Fig. 2a. The gasification temperature range is 500–1000 °C, and the temperature gradient is 50 °C. It is noticed that the gasification temperature has a significant effect on the product gas composition. While temperature is increased from 500 °C to 1000 °C, H_2 content is increased firstly and achieved the maximal point of 18.79% at 700 °C, and then decreased to 16.51% at 1000 °C. CO content is increased all the way to 24.05% at 1000 °C, with a dramatically increase between 500 °C and 650 °C. CO_2 and CH_4 contents decreased while temperature went up. This result was similar to the experimental data of hardwood chips gasification in a downdraft gasifier reported by Son et al. [29]. N_2 content was not displayed since N_2 only acted as dilute gas in the subsequent catalytic conversion process [30] and its content could be calculated by subtracting the sum of other components from 100%.

These trends may be explained by the chemical reactions occurred in the gasification process. The Boudouard reaction (R8), water gas reaction (R9) and steam methane reforming reaction (R11) are endothermic reactions, as shown in Table 1. With the temperature rises, the reaction equilibrium will shift from the reactants to the products. Thus, the reactions between char, CO_2 and H_2O will be enhanced, more CO_2 will be consumed while more CO will be produced. The reaction between CH_4 and H_2O will also be strengthened, with more CH_4 consumed and more CO produced. The water gas shift reaction (R7) and methanation reaction (R10) is exothermic reaction, higher temperature will hinder the reaction and create less CO_2, H_2 and CH_4. The fluctuation of H_2 content might be caused by the combined effects of the reactions in the gasification reduction zone. Water gas shift reaction (R7) has been found one of the most important reactions for tuning the final gas composition during biomass gasification, due to its capability of reacting CO with H_2O to form H_2 and CO_2 [31]. Inflection points in gas composition were observed in Fig. 2a around 600–650 °C, it was proposed that water gas shift reaction (R7) might play a major role in these changes. In lower temperature, water gas shift reaction (R7) was prevailing in producing hydrogen while the reaction was hindered with the temperature increased. Methanation reaction (R10) would consume less H_2 in higher temperature because it was exothermal reaction. The other hydrogen contributing reactions R9 and R11 were endothermic reactions. They may contribute to the hydrogen content increase before 600 °C, however, their reactions may be limited after 600 °C due to lack of reactants such as CH_4 and H_2O. The combined effects of R7, R9, R10 and R11 may cause the decrease of hydrogen content after 650 °C.

The effect of gasification temperature on H_2/CO ratio and gas lower heating value was plotted in Fig. 2b. H_2/CO ratio experienced a sharp

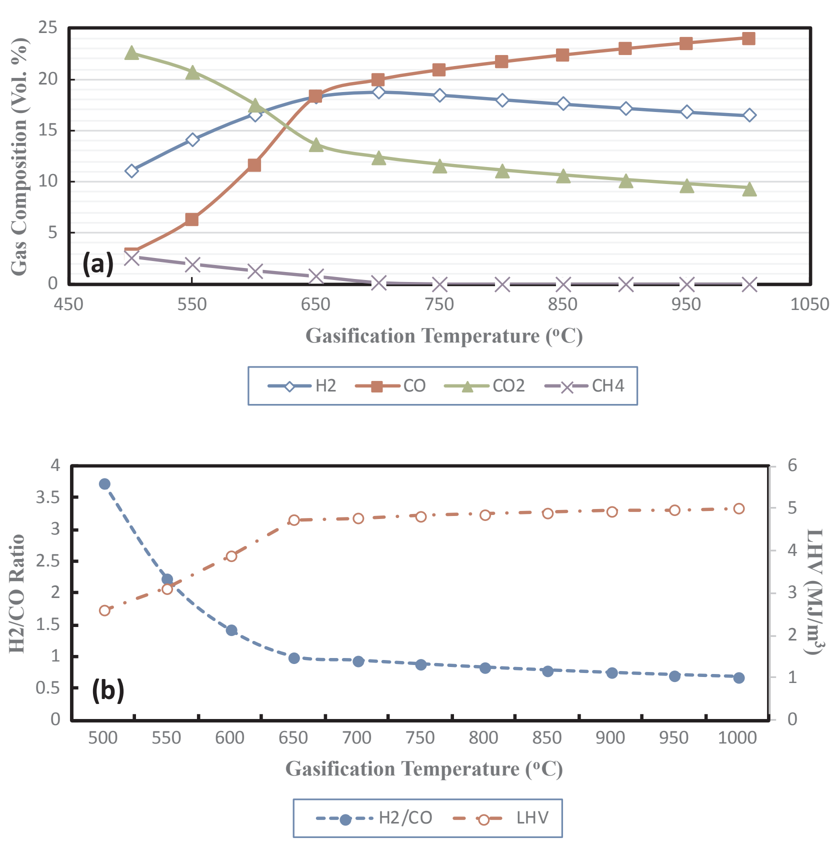

Fig. 2. Effect of gasification temperature on (a) product gas composition and (b) hydrogen to carbon monoxide ratio and lower heating value.

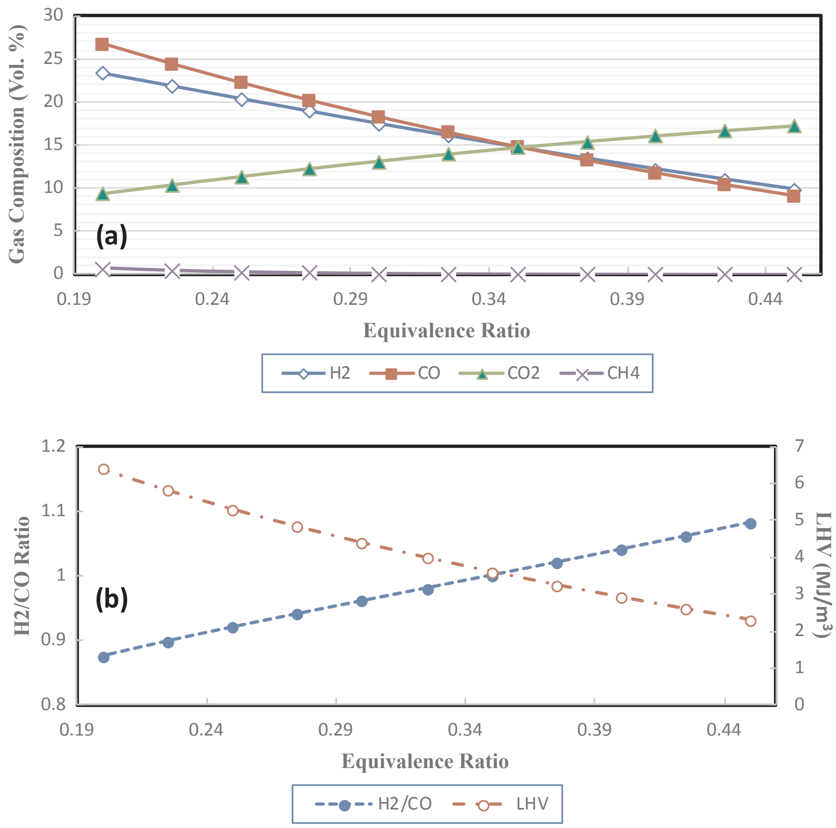

Fig. 3. Effect of equivalence ratio on (a) product gas composition and (b) hydrogen to carbon monoxide ratio and lower heating value.

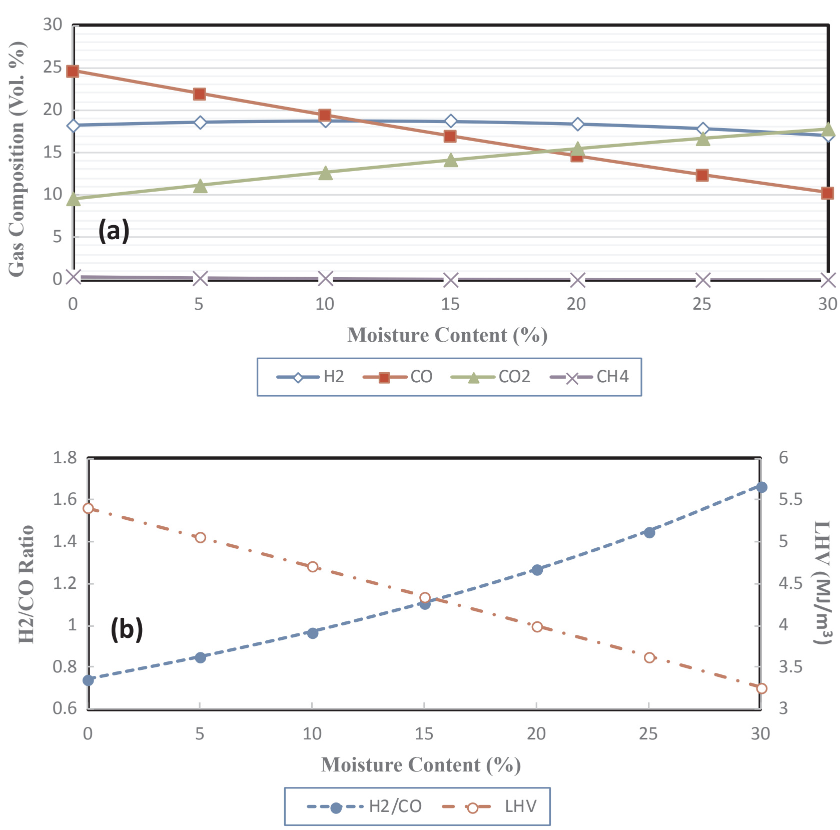

Fig. 4. Effect of biomass moisture content on (a) product gas composition and (b) hydrogen to carbon monoxide ratio and lower heating value.

drop from 3.74 at 500 °C to 1.00 at 650 °C, while went down slowly to 0.68 at 1000 °C. LHV experienced a sharp increase from 2.61 at 500 °C to 4.73 MJ/Nm^3 at 650 °C, while went up slowly to 5.00 MJ/Nm^3 at 1000 °C.

Equivalence ratio (ER) is defined as the ratio of actual air fuel ratio to stoichiometric air fuel ratio, the biomass gasification reaction can happen in the equivalence ratio range from 0.2 to 0.45 [23], and the step of ER is 0.025. The effect of the equivalence ratio on syngas composition was plotted in Fig. 3a. It was observed that H_2 content decreased all the way from 23.40% at 0.2 to 9.83% at 0.45 with the equivalence ratio increased. CO content also decreased all the way from 26.73% at 0.2 to 9.09% at 0.45. CO_2 content increased from 9.40 at 0.2 to 17.21 at 0.45. CH_4 contents decreased while ER went up.

The effect of the equivalence ratio on H_2/CO ratio and syngas LHV was plotted in Fig. 3b. H_2/CO ratio had a linear increase from 0.87 at 0.2 to 1.08 at 0.45. LHV linearly dropped from 6.38 at 0.2 to 2.29 at 0.45. Devi et al. [32] also reported that higher equivalence ratio in biomass gasification would result in decreasing of H_2 and CO contents and LHV, but increasing CO_2 content. Hence, it was recommended that equivalence ratio should be around 0.2 to 0.3 to achieve a good quality bio-syngas for the downstream catalytic synthesis.

The moisture from biomass may also be involved in chemical reactions, such as water gas shift reaction (R7), water gas reaction (R9), and steam methane reforming reaction (R11), and affect the chemical reaction equilibrium and change the component distribution in the syngas product [33]. The moisture content in dry biomass after the pre-drying process was 0–30% [23], so this range was investigated in the sensitivity study. The effect of moisture content on gas composition was plotted and shown in Fig. 4a. It was observed that while the moisture content increased, H_2 content slightly increased from 18.30% at 0% MC to 18.80% at 10% MC, and then slightly decreased to 17.06% at 30% MC. CO content sharply decreased from 24.62% at 0% MC to 9.09% at 30% MC. CO_2 content increased from 9.55% at 0% MC to 17.86% at 30% MC. CH_4 contents decreased while MC went up. The effect of moisture content on H_2/CO ratio and gas LHV was plotted in Fig. 4b. H_2/CO ratio had a linear increase from 0.74 at 0% MC to 1.66 at 30% MC. LHV linearly dropped from 5.41 MJ/Nm^3 at 0% MC to 3.25 MJ/Nm^3 at 30% MC.

5. Conclusions

The following conclusions can be drawn from the previous study and discussion.

- A comprehensive computer model of downdraft biomass gasification process was developed using Aspen Plus based on Gibbs free energy minimization method, modified with restricted chemical reaction equilibrium in the gasification reduction zone.
- The model was successfully validated with experimental data of downdraft hardwood chips gasification, with satisfactory accuracy on the main bio-syngas compositions.
- The effects of gasification temperature, equivalence ratio, and biomass moisture content on bio-syngas quality (gas composition, H_2/CO ratio, and lower heating value) were investigated through the sensitivity analysis.
- All the investigated factors had a significant effect on the bio-syngas quality.
- In order to achieve a good bio-syngas quality for the integrated system operation and downstream catalytic conversion such as Fischer-Tropsch Synthesis, it was recommended that the gasification temperature should be around 650–800 °C, the equivalence ratio should be around 0.2–0.3, and the biomass moisture content should be less than 10%.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.enconman.2017.10.030.

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